

WASKISH, MN, USA

WMO: 727486

Lat: 48.154N

Lon: 94.517W

Elev: 1181

StdP: 14.08

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 04983

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.0	-17.8	-28.9	1.1	-24.8	-24.0	1.5	-17.4	19.7	14.6	18.1	14.5	3.8	240	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	84.2	72.1	81.7	69.7	79.4	67.5	74.2	82.2	71.9	79.5	69.8	77.3	7.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.9	123.3	80.7	69.9	114.8	78.3	66.4	101.4	74.7	38.5	81.9	36.4	78.6	34.5	78.0	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.7	17.4	15.7	-30.4	91.0	6.4	5.2	-35.0	94.7	-38.7	97.7	-42.4	100.7	-47.0	104.4	
			-31.1	77.3	6.6	1.9	-35.9	78.7	-39.7	79.8	-43.4	80.8	-48.2	82.2	
			DB												
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.5	7.7	9.9	25.2	39.5	52.0	62.4	67.3	64.6	57.0	43.3	28.8	14.1
	DBStd	23.37	15.25	13.50	13.96	9.90	8.71	6.30	5.60	6.07	8.33	8.82	11.16	12.75
	HDD50	5645	1310	1122	773	332	79	2	0	0	27	249	638	1113
	HDD65	9575	1775	1542	1234	764	413	121	41	82	264	675	1086	1578
	CDD50	1803	0	0	4	18	141	372	537	453	235	41	2	0
	CDD65	260	0	0	0	0	9	42	114	70	23	2	0	0
	CDH74	2272	0	0	0	5	128	372	940	609	197	21	0	0
	CDH80	442	0	0	0	0	26	60	197	122	34	3	0	0
Wind	WSAvg	6.5	6.3	6.2	6.4	7.1	7.1	6.4	5.9	5.5	6.4	7.3	7.2	6.3
Precipitation	PrecAvg	25.8	0.7	0.6	1.1	1.6	3.0	4.3	3.9	3.3	2.9	2.3	1.3	0.9
	PrecMax	31.4	1.8	1.2	3.0	3.6	6.2	10.9	6.6	6.7	6.9	4.4	3.4	1.9
	PrecMin	18.0	0.2	0.0	0.3	0.7	1.7	1.4	1.5	1.3	0.6	0.3	0.2	0.3
	PrecStd	4.1	0.4	0.3	0.7	0.6	1.3	2.1	1.3	1.4	1.6	1.1	0.8	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.5	45.2	65.9	72.7	83.9	85.7	88.4	88.1	83.9	77.2	61.0	42.6
		MCWB	36.2	39.4	55.5	57.2	67.1	70.9	75.1	74.2	72.2	60.7	51.3	38.4
	2%	DB	36.2	37.3	54.9	65.9	77.4	81.4	84.2	82.5	80.5	69.5	54.1	37.0
		MCWB	34.1	33.9	47.2	52.3	61.6	68.4	73.1	70.8	68.3	58.2	48.6	34.7
	5%	DB	33.6	34.4	47.9	61.4	72.7	78.9	81.7	80.9	75.0	63.0	48.0	34.2
		MCWB	32.1	32.3	41.8	48.5	59.0	66.2	70.7	69.5	65.9	54.3	43.5	32.8
	10%	DB	28.4	30.1	43.1	56.7	68.5	75.0	79.3	77.4	72.0	56.9	44.7	31.5
		MCWB	27.4	29.0	38.1	46.3	57.1	64.6	68.4	67.1	64.0	50.3	40.9	30.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.3	39.7	58.7	58.4	68.9	74.8	77.2	75.5	73.6	63.0	54.1	38.8
		MCDB	39.2	45.3	62.9	69.4	79.1	82.4	84.9	84.2	82.0	70.8	57.5	40.8
	2%	WB	34.0	35.0	47.6	53.2	64.8	71.0	74.7	73.3	70.1	59.2	48.0	35.3
		MCDB	35.7	37.6	55.2	63.1	73.6	78.8	82.6	81.1	77.2	68.6	52.9	37.1
	5%	WB	31.1	32.2	42.3	50.1	61.5	68.2	72.5	70.9	67.2	54.7	44.2	32.8
		MCDB	32.6	34.5	47.4	59.8	69.9	75.6	79.9	78.3	73.5	60.4	47.5	33.9
	10%	WB	28.0	28.8	38.3	47.2	58.7	66.3	70.3	68.7	64.5	51.7	40.5	30.1
		MCDB	28.5	30.1	42.4	56.4	66.5	73.2	77.7	75.9	69.9	56.5	44.5	31.1
Mean Daily Temperature Range	5% DB	MDBR	18.1	20.9	21.2	21.2	21.5	19.9	20.1	21.1	19.9	16.3	13.9	15.1
		MCDBR	16.1	19.9	24.6	29.6	28.6	23.9	22.4	23.9	24.5	25.0	19.9	15.7
		MCWBR	14.0	16.1	16.6	17.7	15.6	13.2	13.2	13.7	14.9	15.5	14.0	13.6
	5% WB	MCDBR	15.2	18.2	22.4	26.6	23.9	20.1	20.5	21.2	20.9	21.4	18.4	14.9
		MCWBR	13.4	14.9	15.7	17.1	14.7	12.7	13.4	13.0	14.0	14.0	13.6	13.2
Clear-Sky Solar Irradiance	taub		0.254	0.277	0.306	0.352	0.363	0.375	0.382	0.377	0.350	0.306	0.279	0.257
	taud		2.361	2.306	2.388	2.366	2.392	2.381	2.382	2.418	2.473	2.576	2.510	2.397
	Ebn at Noon		265	281	290	285	285	281	277	274	272	271	255	249
	Edh at Noon		21	29	31	36	36	37	36	34	29	22	19	18
All-Sky Solar Radiation	RadAvg		468	848	1220	1525	1702	1812	1946	1630	1204	720	420	337
	RadStd		51	72	123	133	102	137	89	120	91	109	46	37

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page